The Blind Spot

Session I · 7 September 2026 — the trial that could not answer, the study that could not see

Class notes · an automated summary of the live session, structured for study. Where these notes and the assigned papers disagree, the papers win.

Key outcomes

First session of the 14-session elective "EBM Through Blind Spots: How to Study Epidemics and Addictions" (Faculty of Liberal Arts and Sciences). The central thesis: even well-executed, peer-reviewed research contains systematic blind spots — information that cannot be seen from the published article itself — and there are concrete tools for identifying and calculating what is missing. Three extended case studies carried the evening: the Lindsay Clancy filicide trial (postpartum psychosis vs. depression), the Pfizer mRNA COVID-19 vaccine trial (rare side effects and unmeasured outcomes), and a Massachusetts retrospective cohort on opioid agonist therapy (confounding by self-selection). The course is taught in English and assessed by two in-class controls (30% each) and an essay (40%).

Course identity

- Format: 14 sessions, elective, open to all bachelor students; designed to be useful not only for future medics but for media workers, journalists, sociologists — anyone who works with evidence
- Instructor: Oksana Stanevich — licensed physician, infectious diseases; doctoral research in Israel on HIV migration dynamics between Russia and Israel, largely computational
- Course angle: the opposite perspective from a standard EBM introduction — not what studies reveal, but what they structurally cannot see: design limits, instrument limits, missing definitions

The central concept: the blind spot

- The metaphor is from human vision: the visual system fills the gap so seamlessly that the viewer never perceives it — researchers likewise do not notice what their study cannot see
- Where blind spots come from:
  - Missing definitions — the case definition may be unstated or contested, making the study population ambiguous
  - Instrument limits — a perfectly executed study is still bounded by what its measurement tools can register
  - Underpowered samples — if the group is too small to statistically encounter a rare event, the event cannot appear in the data, however good the design
  - Unmeasured outcomes — if an outcome was never an endpoint, the study holds no data on it; absence of evidence is not evidence of absence
  - Confounding — a measured effect may be driven in part by an unmeasured third variable

Case 1 — the Lindsay Clancy trial

- Facts: a mother of three killed her children and attempted suicide; after more than 70 witnesses, the jury declared itself unable to reach a verdict
- Legal crux: the defence argued postpartum psychosis (dissociation, no deliberate intent); the prosecution argued postpartum depression (deliberate action)
- Medical complexity: more than ten psychiatric medications prescribed simultaneously by several doctors, without centralised supervision

The guideline conflict

- DSM-5: postpartum onset specifier covers 4 weeks after birth
- ICD-11: about 6 weeks
- A 2025 consensus in Biological Psychiatry: proposes 12 weeks
- The crime occurred roughly a year after birth — outside every published window, including the newest one; yet the jury still could not assign guilt

What the blind spot reveals

- DSM and ICD were built to guide treatment, not to assign legal responsibility — using them as forensic instruments is a category error
- No court-specific criteria for distinguishing culpability from clinical condition existed; every tool in the courtroom had been designed for a different purpose
- Whose fault? Doctors, industry, the insurance system, the court — the honest answer is that no single failure can be named; specific things were simply absent. That absence is what a blind spot looks like in practice

Interlude — bite-mark evidence

- For decades, bite-mark impressions were used in forensic practice as individualising evidence, by analogy with fingerprints
- In 2009 the National Academy of Sciences (USA) found the method scientifically unvalidated; board-certified odontologists shown photographs often could not agree on whether a mark was a bite at all
- At least 26 people are documented as wrongfully convicted, arrested or charged on bite-mark evidence
- Lesson: an instrument can be institutionally accepted for decades while remaining unvalidated — it arrived with confidence attached and no statement of its field of vision

Case 2 — the Pfizer mRNA vaccine trial

- The trial (Polack et al., NEJM 2020) had about 21,700 vaccinated participants; young males 16–19 were assumed in class to be ~10% of them (~2,000)
- Declared outcomes: symptomatic infection and severe disease; not asymptomatic infection, not transmission
- Later Israeli data (Mevorach et al., NEJM 2021): myocarditis after the second dose in young males at ~15 per 100,000

The calculation done in class

- 2,000 young males × 15/100,000 = 0.3 expected cases in the whole trial
- The study could not have registered even one case — it had no statistical power to see this event; to expect a single case it would have needed roughly three times more young male participants
- Two blind spots: (1) rare adverse events were arithmetically invisible at this sample size; (2) transmission was never an outcome
- The media layer then reported "possible heart damage in young males" without the denominator or the arithmetic — and the public read the gap as concealment

Case 3 — opioid agonist therapy cohort (Massachusetts)

- Retrospective cohort, N = 17,568, twelve months after a non-fatal overdose; groups: methadone, buprenorphine, no medication
- Findings: methadone associated with ~53% lower all-cause mortality; buprenorphine ~37%
- Headline: "Addiction medication after an overdose cuts the risk of dying by more than half"

The confounding problem

- Methadone requires attending a clinic every day — the people able and willing to do that are already more organised and more health-engaged than those who are not
- The mortality reduction therefore mixes the pharmacology with the discipline of those who chose it, and a retrospective design cannot separate the two
- The confounder (motivation, stability) was not measured — and the headline cannot carry the caveat
- Important framing from class: this is one of the best papers in the field — and it still cannot answer the causal question the headline answered

Shulgin and the regulation of psychoactive research

- Alexander Shulgin, chemist and psychopharmacologist (died 2014), resynthesised MDMA and created dozens of phenethylamines and tryptamines
- His evidence base: a self-selected group of about twelve people, rating effects on his own scale — and for many compounds that remains the best human data in existence, because the alternative was made illegal
- The books PiHKAL and TiHKAL document the compounds; the archive is at UC Berkeley
- The regulatory parallel to the Clancy case: what guidelines permit and what courts or clinics need are not the same thing

Assessment

- Interim control (week VIII): 30% — exemplary cohort papers studied in advance; an unseen paper of comparable design on the day
- Final control (week XIV): 30% — same mechanics, with a systematic review as the exemplar
- Essay: 40% — 1500 words, topic agreed with the teacher

Vocabulary introduced

Term	As introduced
Blind spot	Information structurally absent from a study, invisible even in a well-executed article
Outcome	The endpoint a study is designed to measure; unmeasured outcomes produce no data
Statistical power	The ability of a sample size to detect an event of a given frequency
Confounding	A third variable that distorts the apparent effect of exposure on outcome
Selection bias	Systematic difference between who enters a study and the wider population
Retrospective cohort	A group observed backwards in time through existing records
Peer review	Pre-publication revision; the pre-review manuscript is not public

Check before you lean on it

- The automated summary called this a 12-session course; the course information sheet says 14 sessions — fixed above.
- The 2009 bite-mark report is by the US National Academy of Sciences, not a British body.

- The vaccine trial is Polack et al., NEJM 2020; the myocarditis frequency is from Mevorach et al., NEJM 2021 (the transcript garbled the names into one study).
- Shulgin's books are PiHKAL and TiHKAL — two volumes, not one.